

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA1610

COURSE NAME: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.

Assessment Tool Weightage: 40%

QUESTIONS: 05

Total Marks:50

Annexure A

Industry Problem Solving Task

<i>Criteria</i>	Understanding of Industry Problem (2)	Application of Engineering Concepts (2.5)	Solution Design (2.5)	Use of Tools (2)	Analysis (1)	<i>Total(10)</i>
<i>Weightage</i>	20%	25%	25%	20%	10%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I TEJASWINI M 192411079 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Tejaswini.M

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Annexure A

Q1 . Dataset: Supermarket Purchases

Customer	Visits/Month	Avg Spend (₹)	Items Bought
C1	10	2000	15
C2	4	5000	8
C3	12	1800	18
C4	3	6000	6
C5	11	2100	16

Question:

Using the above dataset, propose a clustering approach to segment customers and explain how it supports targeted marketing. (BL3 – 7 Marks)

Answer

A suitable technique for this dataset is **K-Means clustering**.

The three features used are:

- Number of visits per month
- Average spending
- Number of items purchased

Since these features have different numerical ranges, **normalization should be performed before clustering**.

Proposed Clusters

Based on the data, we can identify two main customer groups.

Cluster 1 – Frequent, Low-Spending Customers

- C1 → 10 visits, ₹2,000, 15 items
- C3 → 12 visits, ₹1,800, 18 items
- C5 → 11 visits, ₹2,100, 16 items

These customers visit frequently and purchase many items but have relatively low average spending.

Cluster 2 – Less Frequent, High-Spending Customers

- C2 → 4 visits, ₹5,000, 8 items
- C4 → 3 visits, ₹6,000, 6 items

These customers visit less frequently but have a high average spending amount.

Targeted Marketing

For Cluster 1:

- Offer loyalty points.
- Provide frequent-visit rewards.
- Give discounts on bulk purchases.
- Send personalized coupons.

For Cluster 2:

- Promote premium products.
- Provide high-value membership programs.
- Offer personalized premium deals.
- Encourage more frequent visits.

Conclusion

K-Means clustering can divide supermarket customers according to their purchasing behavior. The resulting customer segments allow the supermarket to develop **personalized marketing strategies**, improve customer engagement, and increase sales.

Q2. Dataset: Credit Card Transactions

Transaction Amount (₹) Location Time (hrs)

T1	200	Chennai	10
T2	5000	Delhi	2
T3	150	Chennai	11
T4	7000	Mumbai	1
T5	180	Chennai	12

Question:

Design a clustering-based solution using this dataset to detect unusual or fraudulent transactions. (BL4 – 7 Marks)

Answer

A suitable approach is **clustering-based anomaly detection**.

The transaction features are:

- Transaction amount
- Location
- Transaction time

The location is categorical, so it must be converted into a suitable numerical representation before applying a distance-based clustering algorithm.

Proposed Approach

Step 1 – Data preprocessing

- Handle categorical location values.
- Normalize transaction amount and time.
- Prepare the data for clustering.

Step 2 – Apply clustering

K-Means or another clustering technique can be used to identify normal transaction patterns.

For example, T1, T3 and T5 show a similar pattern:

- Low transaction amounts
- Chennai location
- Late-morning transactions

T2 and T4 are significantly different because they involve:

- Much higher transaction amounts
- Different locations
- Very early transaction times

Possible Identification

Transaction Observation

T1	Normal pattern
T2	Potentially unusual
T3	Normal pattern
T4	Potentially unusual
T5	Normal pattern

T2 and T4 could therefore be flagged for further investigation.

Important: This dataset alone does not prove that T2 or T4 is fraudulent. They are only **anomalous based on the supplied features**.

Fraud Detection Process

Transaction Data



Data Preprocessing



Feature Encoding & Normalization



Clustering



Identify Normal Patterns



Detect Distant/Unusual Transactions



Flag for Investigation

Conclusion

Clustering can identify unusual transaction patterns without requiring a fully labeled fraud dataset. Transactions that are significantly different from normal clusters can be flagged for additional verification, helping improve fraud detection.

Q3. Dataset: Telecom Usage

User Call Duration (min) Data Usage (GB) Recharge (₹)

U1	300	5	200
U2	100	2	100
U3	350	6	220
U4	80	1	90
U5	320	5.5	210

Question:

Explain how clustering can be applied to identify churn-prone users and improve customer retention strategies. (BL4 – 7 Marks)

Answer

Clustering can group telecom users according to their usage and recharge behavior.

The features are:

- Call duration
- Data usage
- Recharge amount

A suitable algorithm is **K-Means clustering** after normalization.

Possible Customer Segments

Cluster 1 – High-Usage Users

- U1
- U3
- U5

These users have:

- High call duration
- High data usage

- Higher recharge amounts

Cluster 2 – Low-Usage Users

- U2
- U4

These users have:

- Low call duration
- Low data usage
- Lower recharge amounts

Churn Analysis

Low-usage customers may represent a potential churn-prone segment if low usage indicates declining engagement.

For example:

- U4 has only 80 minutes of calls.
- U4 uses only 1 GB of data.
- U4 recharges ₹90.

Such customers could be monitored more closely.

Retention Strategies

For low-engagement customers:

- Provide special recharge offers.
- Offer additional data benefits.
- Send personalized discounts.
- Provide loyalty rewards.
- Recommend suitable plans.

For high-usage customers:

- Offer premium plans.
- Provide loyalty benefits.
- Give additional data or calling benefits.

Conclusion

Clustering helps telecom companies divide customers according to usage behavior. Identifying low-engagement groups allows the company to create targeted retention campaigns and potentially reduce customer churn.

Q4 . Dataset: E-commerce Behavior

User Product Views Clicks Purchases

U1	50	20	5
U2	30	10	2
U3	60	25	6
U4	20	8	1
U5	55	22	5

Question:

Suggest a clustering technique to group users and explain how it improves recommendation systems. (BL3 – 7 Marks)

Answer

K-Means clustering is a suitable technique for grouping users based on their online behavior.

The features are:

- Product views
- Number of clicks
- Number of purchases

Possible Clusters

Cluster 1 – Highly Engaged Users

- U1
- U3
- U5

These users have:

- High product views
- High clicks
- High purchases

Cluster 2 – Less Engaged Users

- U2
- U4

These users have lower activity and fewer purchases.

Recommendation Process

The clustering result can be used to recommend products based on group behavior.

For example:

Highly engaged users

- Recommend premium products.
- Recommend related products.
- Provide personalized product suggestions.
- Use cross-selling and upselling.

Less engaged users

- Recommend popular products.
- Provide discounts.
- Show simple personalized suggestions.
- Display products related to previously viewed items.

Architecture

User Behavior Data



Data Preprocessing



Feature Normalization



K-Means Clustering



User Segmentation



Analyze Cluster Preferences



Personalized Recommendations

Conclusion

Clustering helps recommendation systems identify users with similar behavior. Instead of giving the same recommendations to every user, the system can provide **cluster-specific and personalized recommendations**, improving user experience and potentially increasing purchases.

Q5. Dataset: Traffic Data

Location	Vehicle Count	Avg Speed (km/h)	Delay (min)
L1	200	30	10
L2	500	15	25
L3	220	28	12
L4	600	10	30
L5	210	32	9

Question:

Design a clustering-based solution to identify traffic congestion zones and explain how it helps in traffic management. (BL4 – 7 Marks)

Answer

A suitable approach is **K-Means clustering** using:

- Vehicle count
- Average speed
- Delay

These features should be normalized because they have different scales.

Possible Clusters

Cluster 1 – High Congestion

- L2
- L4

Characteristics:

- High vehicle count
- Low average speed
- High delay

L4 is especially congested because it has:

- 600 vehicles
- 10 km/h average speed
- 30 minutes delay

Cluster 2 – Low Congestion

- L1
- L3
- L5

These locations have:

- Lower vehicle counts
- Higher average speeds
- Lower delays

Traffic Management

For high-congestion zones:

- Adjust traffic signal timing.
- Deploy traffic police.
- Suggest alternative routes.
- Increase road monitoring.
- Provide real-time traffic alerts.

For low-congestion zones:

- Continue regular monitoring.
- Use these roads as possible alternative routes.

Proposed System

Traffic Data



Data Cleaning



Feature Normalization



K-Means Clustering



Traffic Zone Segmentation



Identify Congestion Zones



Traffic Management Actions

Conclusion

Clustering can divide locations into different congestion levels based on vehicle count, speed, and delay. High-congestion clusters can be prioritized for traffic management actions, helping authorities reduce delays and improve road efficiency.